Study notes: Convex Optimization, Module I

Binmore, *Mathematical Analysis: A Straightforward Approach* (2nd ed., 1982), Chapter 19, Vector derivatives. Numbers are the book's own. Figures are described in words, not drawn yet. Exercises are not included in the notes. Student notes in my own words; every numbered item is kept with its book number.

Section 1: Directional and partial derivatives

Binmore 2nd ed., printed pp. 190–195 (19.1–19.5)

Before you start

- A **unit vector** has length 1, so $\|\mathbf{u}\| = 1$. A vector (a, b) has length $\sqrt{a^2 + b^2}$.
- The **derivative** of a function of one variable is the limit $\lim_{t \to 0} \dfrac{g(t) - g(0)}{t}$, when it exists (Binmore §10.1).
- A **function of two variables** $f(x_1, x_2)$ has a graph, the surface $y = f(x_1, x_2)$, over the plane.

Section 1. Directional and partial derivatives

19.1 Directional derivatives

Take two vectors $\boldsymbol{\xi}$ and $\mathbf{u}$ in $\mathbb{R}^n$. Read $\boldsymbol{\xi}$ as a point and $\mathbf{u}$ as a direction, with $\|\mathbf{u}\| = 1$. The points $\mathbf{x} = \boldsymbol{\xi} + t\mathbf{u}$, for real t, form the straight line through $\boldsymbol{\xi}$ in the direction $\mathbf{u}$. The number t is the distance from $\boldsymbol{\xi}$ to $\mathbf{x}$ along that line.

Let $f : \mathbb{R}^n \to \mathbb{R}$, and define $F(t) = f(\boldsymbol{\xi} + t\mathbf{u})$. For $n = 2$ we can see what F means. Cut the surface $y = f(\mathbf{x})$ by a vertical plane through $\boldsymbol{\xi}$ that is parallel to $\mathbf{u}$. The cut is the curve $y = F(t)$, with $\boldsymbol{\xi}$ at $t = 0$ [figure described in the book].

The slope of that cut at $t = 0$ is $F'(0)$. It is the rate at which the surface rises at $\boldsymbol{\xi}$ when we move in the direction $\mathbf{u}$.

> **DIRECTIONAL DERIVATIVE**
> The directional derivative of f at $\boldsymbol{\xi}$ in the direction $\mathbf{u}$ is written $D_{\mathbf{u}} f(\boldsymbol{\xi})$, and
>
> $$D_{\mathbf{u}} f(\boldsymbol{\xi}) = \lim_{t \to 0} \frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t}.$$

The same symbol is used when $\mathbf{u}$ is not a unit vector, or when f has vector values. The slope picture then no longer applies, but the limit still makes sense.

19.2 Example

Let $f(x_1, x_2) = x_1 x_2$. We want the slope of the surface at $(1, 2)$ in the direction $(3, 4)$.

The vector $(3, 4)$ has length 5, so first replace it by the unit vector $\mathbf{u} = (3/5, 4/5)$, which points the same way. With $\boldsymbol{\xi} = (1, 2)$:

$$\frac{f(\boldsymbol{\xi} + t\mathbf{u}) - f(\boldsymbol{\xi})}{t} = \frac{(1 + 3t/5)(2 + 4t/5) - 2}{t}$$

$$= \frac{2t + 12t^2/25}{t} = 2 + \frac{12t}{25}.$$

> **RESULT**
> As $t \to 0$ this tends to 2. So the slope of $y = x_1 x_2$ at $(1, 2)$ in the direction $(3, 4)$ is 2.

The curve $x_1 x_2 = 2$ is a *contour* of f. Every point of the surface above this contour is at the same height. The direction $(1, -2)$ is tangent to the contour, and the book asks you to check that the directional derivative at $(1, 2)$ in that direction is zero.

19.3 Partial derivatives

Let $f : \mathbb{R}^n \to \mathbb{R}$. To keep the notation short, take $n = 2$. The general case is the same. For $\boldsymbol{\xi} = (\xi_1, \xi_2)$, the *partial derivatives* of f at $\boldsymbol{\xi}$ are defined when the limits exist:

> **PARTIAL DERIVATIVES**
>
> $$f_1(\xi_1, \xi_2) = \lim_{h \to 0} \frac{f(\xi_1 + h, \xi_2) - f(\xi_1, \xi_2)}{h}.$$
>
> $$f_2(\xi_1, \xi_2) = \lim_{k \to 0} \frac{f(\xi_1, \xi_2 + k) - f(\xi_1, \xi_2)}{k}.$$

In practice you do not need the limits. To find f_1, differentiate with respect to x_1 and treat x_2 as a constant. To find f_2, do the reverse. This is ordinary differentiation in one variable at a time.

19.4 Examples

(i) Let $f(x_1, x_2) = x_1^2 x_2 + x_2^3 x_1$. Then $f_1 = 2x_1 x_2 + x_2^3$ and $f_2 = x_1^2 + 3x_2^2 x_1$.

(ii) Let $g(x_1, x_2) = \sin(x_1 x_2 + x_2^2)$. Then $g_1 = x_2 \cos(x_1 x_2 + x_2^2)$ and

$g_2 = (x_1 + 2x_2)\cos(x_1x_2 + x_2^2)$.

(iii) Let $h(x_1, x_2, x_3) = x_1 + x_2x_3$, a function on $\mathbb{R}^3$. Then $h_1 = 1$, $h_2 = x_3$ and $h_3 = x_2$.

The geometric meaning: let $\mathbf{i} = (1, 0)$ and $\mathbf{j} = (0, 1)$, the unit vectors along the two axes. Taking $t \to 0$ in the definition of the directional derivative gives

$$D_{\mathbf{i}}f(\boldsymbol{\xi}) = f_1(\xi_1, \xi_2), \qquad D_{\mathbf{j}}f(\boldsymbol{\xi}) = f_2(\xi_1, \xi_2).$$

So $f_1(\xi_1, \xi_2)$ is the slope of the surface at $\boldsymbol{\xi}$ in the x_1-direction, and $f_2(\xi_1, \xi_2)$ is the slope in the x_2-direction. Cut the surface by the plane $x_2 = \xi_2$: the graph $y = f(x_1, \xi_2)$ on that plane has slope $f_1(\xi_1, \xi_2)$ at $x_1 = \xi_1$ [figures described in the book].

19.5 Notation

A partial derivative has several notations, and you need to read all of them. The book writes $f_{x_1}(\xi_1, \xi_2)$ for $f_1(\xi_1, \xi_2)$, and similarly for f_2. When $y = f(x_1, x_2)$, it writes $\dfrac{\partial y}{\partial x_1}$ for $f_1(x_1, x_2)$ and $\dfrac{\partial y}{\partial x_2}$ for $f_2(x_1, x_2)$.

The $\partial y/\partial x_1$ form is easy to misread. The letters x_1 and x_2 serve both as the point where we evaluate and as the variables we differentiate with respect to. This does not matter in examples, but it causes trouble in proofs.

When several variables are present, say which one is held fixed: $\left(\dfrac{\partial y}{\partial x_1}\right)_{x_2}$ means differentiate with respect to x_1, treating x_2 as a constant.

What to take from this section

- A directional derivative is the slope of the surface $y = f(\mathbf{x})$ at a point, in a chosen direction $\mathbf{u}$. Use a unit vector for $\mathbf{u}$ [19.1, 19.2].
- The partial derivatives f_1 and f_2 are the directional derivatives along the two axes [19.3, 19.4].
- To compute a partial derivative, differentiate in one variable and treat the others as constants [19.3, 19.4].
- Check the length of the direction vector first. If it is not 1, normalise it [19.2].